Hybrid Dynamical Systems

Pantelis Sopasakis

School of Electronics, Electrical Engineering & Computer Science
Centre for Intelligent and Autonomous Manufacturing Systems (i-AMS)
Queen's University Belfast, United Kingdom

Outline

- Hybrid systems: examples
- Hybrid time, arcs, solutions
- Stability notions
- Lyapunov stability theorems

Reading material

Main reference: R. Goebel, R.G. Sanfelice, A.R. Teel, Hybrid dynamical systems: modelling, stability, and robustness, Princeton University Press, 2012.

R.G. Sanfelice, Hybrid Feedback Control, Princeton University Press, 2021.

I. Hybrid systems

Hybrid systems

Roughly: systems where continuous-time and discrete-time behaviours coexist.

Continuous time: ODEs (flows). Discrete time: jumps, instantaneous changes.

Example 1: Bouncing ball

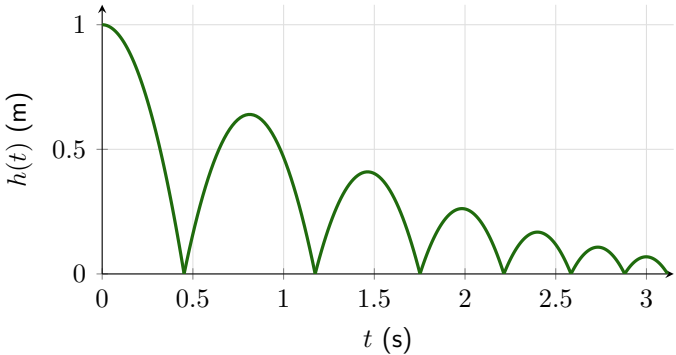
A bouncing ball can be modelled as

$$\begin{bmatrix} \dot{h} \\ \dot{v} \end{bmatrix} = \begin{bmatrix} v \\ -g \end{bmatrix}, \text{ for } h > 0$$
$$\begin{bmatrix} h^+ \\ v^+ \end{bmatrix} = \begin{bmatrix} h \\ -\lambda v \end{bmatrix}, h = 0, v < 0$$

where $h^+ = h(t^+)$ and $\lambda \in [0, 1)$. This is an example of an *impulsive system* with state-dependent jumps.

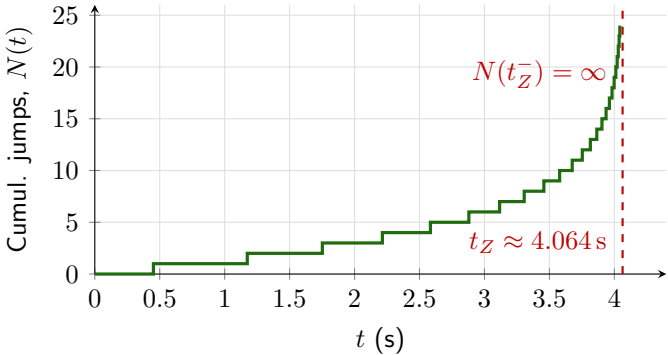
Bouncing ball simulation

Here is a simulation with $g = 9.81\,\mathrm{m/s^2}$, $\lambda = 0.8$, $h(0) = 1$, and $v(0) = 0$.



Bouncing ball: Zeno behaviour

We have an infinite number of bounces in finite time.



Example 2: Drug administration

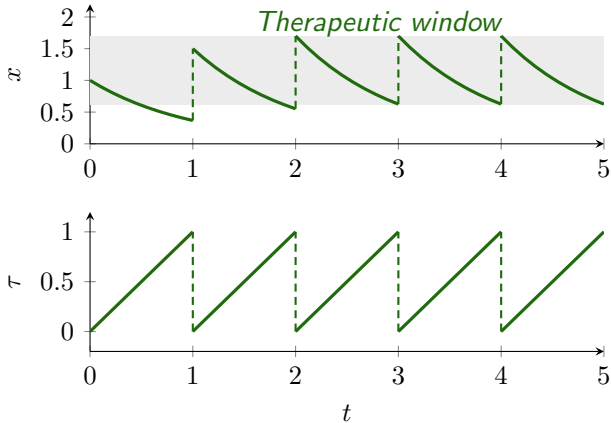
Let $x \in \mathbb{R}^n$ be the amount of a drug in the n compartments (organs/tissues) of the body. Every T the patient takes a pill with amount u of drug, so the amount in the stomach (x_1) jumps. A model of the system is

$$\begin{bmatrix} \dot{x} \\ \dot{\tau} \end{bmatrix} = \begin{bmatrix} f(x) \\ 1 \end{bmatrix}, \text{ for } 0 \leq \tau \leq T$$

$$\begin{bmatrix} x_1^+ \\ x_{2:n}^+ \\ \tau^+ \end{bmatrix} = \begin{bmatrix} x_1 + u \\ x_{2:n} \\ 0 \end{bmatrix}, \tau = T$$

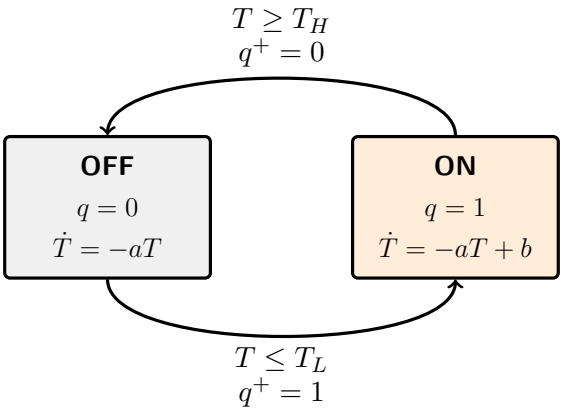
This is an *impulsive system* with time-dependent jumps.

Drug administration: simulations



Example 3: Thermostat

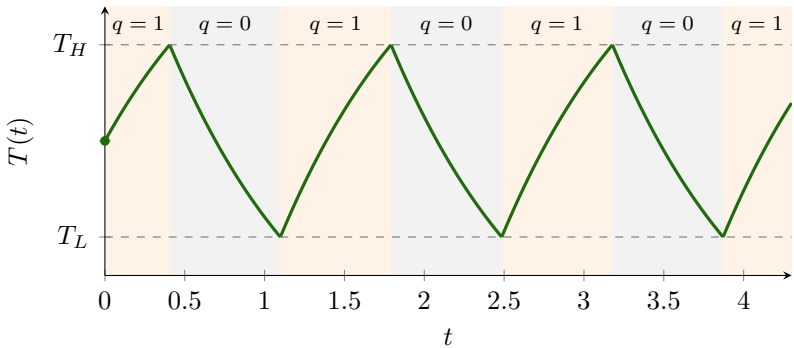
A thermostat is an example of a *switching system*.



Assumptions: $a > 0$ and $0 < T_L < T_H < b/a$

Thermostat: simulations

Simulations with $a = 1$, $b = 3$, $T_L = 1$, $T_H = 2$, and $T(0) = 1.5$.



Example 4: Linear sampled-data system

Control actions, u, are computed by an embedded device at times $t = 0, T_s, 2T_s, \ldots$ and are kept constant over intervals $[jT_s, (j+1)T_s)$. The system dynamics is

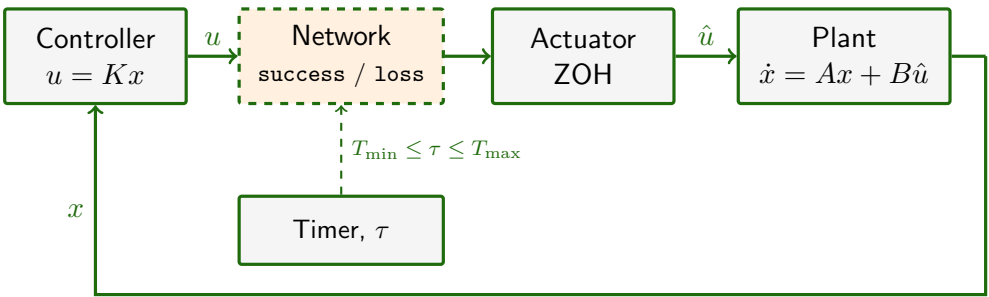
$$\begin{bmatrix} \dot{z} \\ \dot{u} \\ \dot{\tau} \end{bmatrix} = \begin{bmatrix} Az + Bu \\ 0 \\ 1 \end{bmatrix}, \text{ for } \tau \in [0, T_s]$$

$$\begin{bmatrix} z^+ \\ u^+ \\ \tau^+ \end{bmatrix} = \begin{bmatrix} z \\ Kz \\ 0 \end{bmatrix}, \text{ for } \tau = T_s$$

Here τ is a timer ($\dot{\tau} = 1$) which resets whenever $\tau = T_s$.

Example 5: Networked control system

The control actions are transmitted to the actuator (ZOH) over a network. The network permits aperiodic transmissions; inter-transmission times are in $[T_{\min}, T_{\max}]$ and packet dropouts are possible.



Example 5: Networked control system

Let $\xi = (x, \hat{u}, \tau)$, where τ is a network timer. In hybrid-inclusion form,

$$\begin{cases} \dot{\xi} = F(\xi), & \tau \in [0, T_{\max}], \\ \xi^+ \in G(\xi), & \tau \in [T_{\min}, T_{\max}]. \end{cases}$$

where

$$F(\xi) = \begin{bmatrix} Ax + B\hat{u} \\ 0 \\ 1 \end{bmatrix}, G(\xi) = \left\{ \begin{bmatrix} x \\ Kx \\ 0 \end{bmatrix}, \begin{bmatrix} x \\ \hat{u} \\ 0 \end{bmatrix} \right\}.$$

The first value of G represents a successfully received control packet; the second represents packet loss.

Example 6: Event-triggered satellite control

A satellite needs to transmit data to Earth for control actions to be computed. We want to invoke the controller only if the state has changed significantly compared to last time we invoked it.

Let x be the system state, $\hat{x}$ be the last state transmitted to the controller, and let u denote the input. Let $\sigma > 0$. The system state is $z = (x, \hat{x})$ and the dynamics is

$$\begin{cases} \begin{cases} \dot{x} = f(x,u) \\ u = \kappa(\hat{x}) \\ \dot{\hat{x}} = 0 \end{cases} & \text{for } \|x - \hat{x}\| \leq \sigma \|x\| \\ \begin{cases} x^+ = x \\ \hat{x}^+ = x \end{cases} & \text{for } \|x - \hat{x}\| \geq \sigma \|x\|, \text{ and } \|x - \hat{x}\| > 0 \end{cases}$$

Assumption: Communication and computation delays are negligible.

Hybrid inclusions

A representation of hybrid systems is

$$\mathcal{H} : \begin{cases} \dot{z} \in F(z), & z \in C \\ z^+ \in G(z), & z \in D \end{cases}$$

where $z \in \mathbb{R}^n$ is the state, $C, D \subseteq \mathbb{R}^n$, and $F : C \rightrightarrows \mathbb{R}^n$ and $G : D \rightrightarrows \mathbb{R}^n$ are set-valued mappings.

We denote $\mathcal{H} = (C, F, D, G)$.

Examples of hybrid systems

The following systems can be cast as hybrid (why?)

1. Car with manual/automatic gearbox
2. Any system that may fail
3. Traffic lights
4. Robot walking and any engineering system involving impacts (tennis, pool, hitting with hammer)
5. Coulomb friction
6. Systems with state/input saturation

Question: Can you think of other examples?

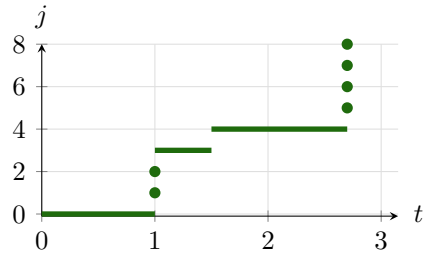
II. Hybrid time, solutions

Hybrid time domain

A **compact hybrid time domain** is of the form

$$E = \bigcup_{j=0}^{J-1} [t_j, t_{j+1}] \times \{j\},$$

with $J \geq 1$, $0 = t_0 \leq t_1 \leq \ldots \leq t_J$. Note that $E \subseteq \mathbb{R}_+ \times \mathbb{N}$.



A set $E \subseteq \mathbb{R}_+ \times \mathbb{N}$ is a **hybrid time domain** if $E \cap ([0, T] \times \{0, \ldots, J\})$ is a compact hybrid time domain whenever $(T, J) \in E$.

Hybrid time domain

Let E be a hybrid time domain. We define $\sup_t E = \sup\{t \geq 0 : \exists j \in \mathbb{N} : (t, j) \in E\}$. Likewise, we define $\sup_j E$. The length of E is $|E| = \sup_t E + \sup_j E$.

For $(t, j), (t', j') \in E$ we define

$$(t, j) \preceq (t', j') \iff t + j \leq t' + j'.$$

Question: Definition of supremum + examples.

Hybrid arc

Let E be a hybrid time domain. A function $\phi : E \to \mathbb{R}^n$ is called a **hybrid arc** if for every j the function $\phi(\,\cdot\,, j)$ is locally absolutely continuous on $I^j = \{t : (t, j) \in E\}$.

An arc is...

1. *Nontrivial* if E contains at least two distinct points
2. *Complete* if $|E| = \infty$
3. *Zeno* if it is complete and $\sup_t E < \infty$
4. *Discrete* if nontrivial and $E \subseteq \{0\} \times \mathbb{N}$
5. *Eventually discrete* if $T := \sup_t E < \infty$ and $E \cap (\{T\} \times \mathbb{N})$ contains ≥ 2 points
6. *Continuous* if nontrivial and $E \subseteq \mathbb{R}_+ \times \{0\}$
7. *Eventually continuous* if $J := \sup_j E < \infty$ and $E \cap (\mathbb{R}_+ \times \{J\})$ contains ≥ 2 points

Question: Definition of (loc.) abs. continuity!

Hybrid solutions

We say $\phi : E \to \mathbb{R}^n$ is a **solution** of $\mathcal{H}(C, F, D, G)$ if $\phi(0,0) \in \overline{C} \cup D$ and

1. $\forall j \in \mathbb{N}$ s.t. I^j has nonempty interior,

$$\phi(t,j) \in C, \forall t \in \operatorname{int} I^j$$
$$\dot{\phi}(t,j) \in F(\phi(t,j)), \text{for almost all } t \in I^j$$

2. $\forall (t,j) \in E$ such that $(t, j+1) \in E$

$$\phi(t,j) \in D,$$
$$\phi(t,j+1) \in G(\phi(t,j)).$$

Questions: (i) Definition of closure of set, (ii) Definition of interior, (iii) Definition of "almost all."

Hybrid solutions: examples

Consider the hybrid system

$$\mathcal{H}_1 : \begin{cases} \dot{x} = 0, & x \in \mathbb{R} \\ x^+ = x + 1, & x \in \mathbb{R} \end{cases}$$

One solution is $\phi_1(t, 0) = 5$ with $\mathbf{dom}\, \phi_1 = \mathbb{R}_+ \times \{0\}$ (continuous). Another solution is $\phi_2(0, j) = j$ with $\mathbf{dom}\, \phi_2 = \{0\} \times \mathbb{N}$ (discrete and Zeno).

Consider the hybrid system

$$\mathcal{H}_2 : \begin{cases} \dot{x} = x^2, & |x| \geq 1 \\ x^+ = x - 1, & x \in [0, 1] \end{cases}$$

Starting from $x_0 = 1$ one solution is $\psi_1(t, 0) = (1 - t)^{-1}$, with $\mathbf{dom}\, \psi_1 = [0, 1) \times \{0\}$ (not complete). Find another solution.

Maximal solutions

A solution ϕ of a hybrid system $\mathcal{H}$ is called **maximal** if there is no other solution ψ with $\mathbf{dom}\,\psi \supsetneq \mathbf{dom}\,\phi$ and $\psi|\,\mathbf{dom}\,\phi = \phi$.

Complete $\implies$ maximal, but maximal $\not\Longrightarrow$ complete. Example: previous slide.

Zeno behaviour of bouncing ball

Recall the bouncing ball model

$$\begin{bmatrix}\dot{h}\\ \dot{v}\end{bmatrix} = \begin{bmatrix} v \\ -g \end{bmatrix}, \text{ for } h > 0$$
$$\begin{bmatrix}h^+\\ v^+\end{bmatrix} = \begin{bmatrix} h \\ -\lambda v \end{bmatrix}, h = 0, v < 0$$

Suppose $\lambda \in (0,1)$. Let $h(0,0) = h_0$ and $v(0,0) = 0$. Then,

$$\begin{aligned} h(t,0) &= h_0 - \tfrac{1}{2} g t^2, \\ v(t,0) &= -gt, \end{aligned}$$

for $t \in [0, t_1]$ where t_1 is the time of the first impact where $h(t_1, 0) = 0$, equiv.

$$t_1 = \sqrt{\frac{2h_0}{g}}.$$

Zeno behaviour of bouncing ball

Let v_1 be the pre-impact velocity, $v_1 = v(t_1, 0) = -gt_1 = -\sqrt{2gh_0}$, so $v_1^+ = \lambda\sqrt{2gh_0}$. Recursively,

$$v_j^+ = \lambda^j \sqrt{2gh_0}.$$

Departing from the ground at velocity v_j^+ the ball reaches the apex after time v_j^+/g, so the next impact is after time

$$\Delta t_j = t_{j+1} - t_j = \frac{2v_j^+}{g} = \frac{2\lambda^j\sqrt{2gh_0}}{g} = 2\lambda^j t_1.$$

The time of the n-th impact is

$$t_n = t_1 + \sum_{j=1}^{n-1} \Delta t_j = t_1 + \sum_{j=1}^{n-1} 2\lambda^j t_1.$$

Notation: The pre-impact velocity is $v(t_j, j-1)$; the post-impact velocity is $v_j^+ = v(t_j, j)$.

Zeno behaviour of bouncing ball

As $n \to \infty$

$$t_n \to t_1 + 2t_1 \sum_{j=1}^{\infty} \lambda^j = t_1 \frac{1+\lambda}{1-\lambda} < \infty,$$

i.e., infinitely many jumps occur before time $t_Z := t_1 \frac{1+\lambda}{1-\lambda}$.

III. Stability notions

Stable + stable = unstable

Switching system with stable modes:

$$\dot{x} = \begin{cases} \begin{bmatrix} -0.9 & 2 \\ 0 & -0.4 \end{bmatrix} x, & q(t) = 1 \\ \begin{bmatrix} -0.5 & 0 \\ 2 & -0.8 \end{bmatrix} x, & q(t) = 2 \end{cases}$$

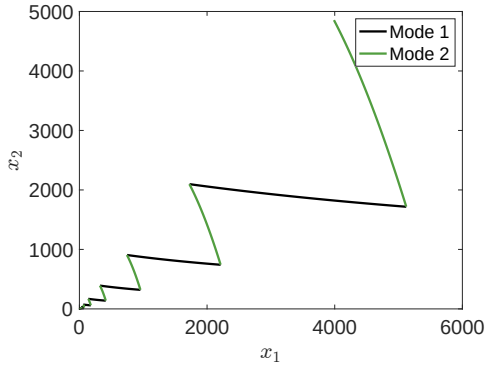
$$\dot{q} = 0, \dot{\tau} = 1$$

whenever $\tau < 0.5$, and jumps

$$\tau^+ = 0, q^+ = 3 - q,$$

when $\tau = 0.5$.

Initial cond. $x(0) = (1, 1)$, $q(0) = 1$, $\tau(0) = 0$.



Unstable + unstable = stable

Switching system with unstable modes:

$$\dot{x} = \begin{cases} \begin{bmatrix} 1 & 0 \\ 1.2 & -3 \end{bmatrix} x, & q(t) = 1 \\ \begin{bmatrix} -3 & 1 \\ 0 & 1 \end{bmatrix} x, & q(t) = 2 \end{cases}$$

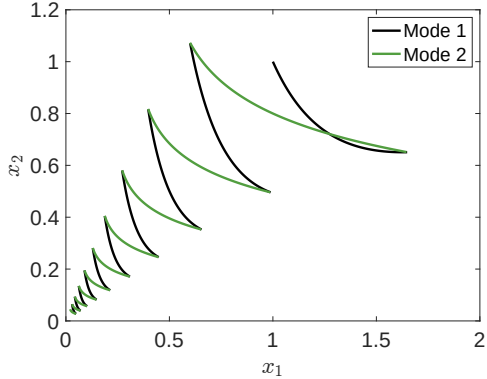
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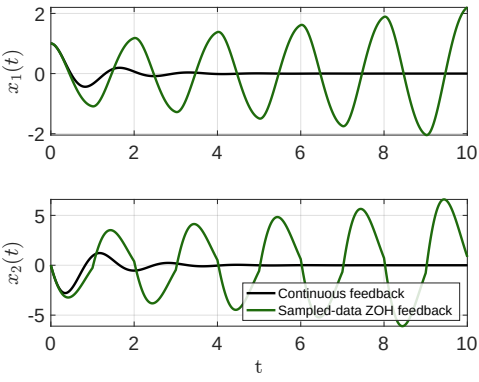


Sampled data becomes unstable

We have the linear system

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -10 & -2 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u,$$
$$u = \begin{bmatrix} -5.5 & 0 \end{bmatrix} x.$$

The closed-loop system is stable, but a sampled-data implementation with a ZOH element and sampling period $T_s = 1$ makes it unstable.



Stability notions

Instead of stability of equilibrium points we have stability of closed sets. We ask

1. **Stability:** Can we control how far the trajectories stray from a set $\mathcal{A}$ using the initial condition?
2. **Pre-attractivity:** if we wait long enough, and provided the trajectories live long enough, will the trajectories go arbitrarily near $\mathcal{A}$?

Stability

Let $\mathcal{A} \subseteq \mathbb{R}^n$ be a nonempty closed set. The distance to $\mathcal{A}$ is defined as

$$\|x\|_{\mathcal{A}} = \inf_{y \in \mathcal{A}} \|x - y\|.$$

We say that $\mathcal{A}$ is *uniformly globally stable* (UGS) for a hybrid system $\mathcal{H}$ if there is $\alpha \in \mathcal{K}_\infty$ such that for every solution ϕ

$$\|\phi(t,j)\|_{\mathcal{A}} \leq \alpha(\|\phi(0,0)\|_{\mathcal{A}}), \forall (t,j) \in \mathbf{dom}\, \phi.$$

Remarks: (i) *uniformly* means that α is a function of the initial condition only—not the particular solution or the time, (ii) *globally* means that $\phi(0,0)$ can be arbitrarily large.

Question: What is a $\mathcal{K}_\infty$-class function?

Remark: UGS = Stability + Uniform global boundedness. Show that UGS implies stability (δ-ϵ definition).

Pre-attractivity

We say that $\mathcal{A}$ is *uniformly globally pre-attractive* (UGpA) if for every $r > 0$ and $\epsilon > 0$ there is a $T > 0$ such that if a solution ϕ satisfies $\|\phi(0,0)\|_{\mathcal{A}} \leq r$, then

$$(t,j) \in \mathbf{dom}\,\phi, t + j \geq T \Longrightarrow \|\phi(t,j)\|_{\mathcal{A}} \leq \epsilon.$$

If all maximal solutions are complete, the prefix "pre-;; is dropped. UGpA is called uniform global attractivity (UGA); for every solution ϕ,

$$\lim_{\substack{(t,j)\in\mathbf{dom}\,\phi \\ t+j\to\infty}} \|\phi(t,j)\|_{\mathcal{A}} = 0.$$

Uniform global pre-asymptotic stability (UGpAS) = UGS + UGpA.

IV. Lyapunov theory

Lyapunov stability theorem

Let $\mathcal{H} = (C, F, D, G)$ be a hybrid inclusion, $\mathcal{A} \subseteq \mathbb{R}^n$ be nonempty & closed, and V be a candidate Lyapunov function for $\mathcal{H}$. If there exist $\alpha_1, \alpha_2 \in \mathcal{K}_\infty$ and a continuous positive definite function ρ such that

$$\alpha_1(\|x\|_{\mathcal{A}}) \leq V(x) \leq \alpha_2(\|x\|_{\mathcal{A}}), \text{ for } x \in C \cup D \cup G(D), \tag{L1}$$

$$\nabla V(x)^\intercal f \leq -\rho(\|x\|_{\mathcal{A}}), \text{ for } x \in C, f \in F(x), \tag{L2}$$

$$V(g) - V(x) \leq -\rho(\|x\|_{\mathcal{A}}), \text{ for } x \in D, g \in G(x), \tag{L3}$$

then $\mathcal{A}$ is UGpAS for $\mathcal{H}$.

Remark: A candidate Lyapunov function is defined on a set that contains $\overline{C} \cup D \cup G(D)$ and is cont. diff. on an open set containing $\overline{C}$.

Recall: Definition of $\mathcal{K}_\infty$-class function. The quantity $\nabla V(x)^\intercal f$ is the time derivative of V along the flow that satisfies $\dot{x} = f(x)$. It is often denoted $L_f V(x)$ (Lie derivative of V along f).

Applications

1. Bouncing ball is UGAS
2. Linear sampled-data systems
3. Switching systems

IV-1. Stability of bouncing ball

Stability of bouncing ball

Recall the bouncing ball model

$$\begin{bmatrix}\dot{h}\\ \dot{v}\end{bmatrix} = \begin{bmatrix} v \\ -g \end{bmatrix}, \text{ for } h > 0$$
$$\begin{bmatrix}h^+\\ v^+\end{bmatrix} = \begin{bmatrix} h \\ -\lambda v \end{bmatrix}, h = 0, v < 0$$

The state is $x = (h, v)$. The total energy is (assume $m = 1$) $E = \frac{1}{2}v^2 + gh$, which satisfies (L1) [w/b]; for (L2) it is

$$\dot{E}(x) = \nabla E(x)^\mathsf{T} f(x) = \begin{bmatrix} g \\ v \end{bmatrix}^\mathsf{T} \begin{bmatrix} v \\ -g \end{bmatrix} = 0,$$

for $x \in C$ ($h > 0$), so (L2) is not satisfied.

Stability of bouncing ball

Regarding Condition (L3),

$$E(g(x)) - E(x) = -\tfrac{1}{2}(1-\lambda^2)\|x\|^2,$$

for $x \in D$ ($h = 0$, $v < 0$), so (L3) is satisfied.

Conclusion: Although E drops with every jump, it remains constant during flows. Close, but no cigar!

Stability of bouncing ball

Let us try to modify E; try

$$V(x) = w(v)E(x),$$

where $w : \mathbb{R} \to \mathbb{R}$ is to be determined. Then, [w/b]

$$\dot{V} = -gw'(v)E(x).$$

To have $\dot{V} < 0$ we need $w'(v) > 0$, i.e., (i) $w \nearrow$ (ii) reasonable assumption: $w(0) = 1$.

First attempt: $w(v) = 1 + \theta v$, for some $\theta > 0$ will not work because $w(v)$ can become negative.

Let's try $w(v) = 1 + \theta \arctan v$ for some $\theta > 0$. Then ...

Stability of bouncing ball

We have [w/b]

$$\dot{V} = -\frac{\theta g}{1+v^2} E(x) < 0,$$

so (L1) is still satisfied [w/b], (L2) is satisfied, and it remains to check (L3). Firstly, $E^+ = \lambda^2 E$ [w/b]. We have

$$\begin{aligned} V(g(x)) - V(x) &= (1 + \theta \arctan(v^+))E^+ - (1 + \theta \arctan(v))E \\ &= \left[(1 + \theta \arctan(-\lambda v))\lambda^2 - (1 + \theta \arctan(v))\right] E \\ &\leq \left[(1 + \theta\tfrac{\pi}{2})\lambda^2 - (1 - \theta\tfrac{\pi}{2})\right] E, \end{aligned}$$

and the expression in the brackets can be made negative by choosing

$$0 < \theta < \frac{2(1-\lambda^2)}{\pi(1+\lambda^2)}.$$

Exercise: Try $w(v) = 1 + \theta \tanh v$.

Stability of bouncing ball

The origin is UGpAS for the bouncing ball system, which is shown using the Lyapunov function

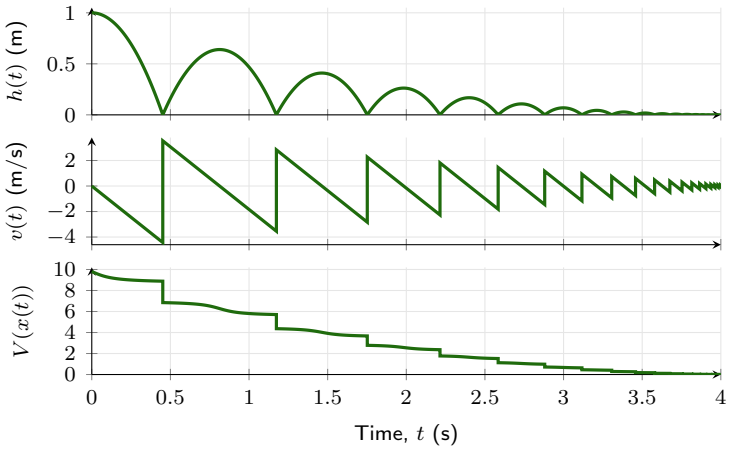
$$V(x) = \left(1 + \frac{(1-\lambda^2)}{\pi(1+\lambda^2)} \arctan v\right) \left(\tfrac{1}{2}v^2 + gh\right).$$

If all maximal solutions are complete, then we have UGAS.

Question: Under what conditions are all maximal solutions complete? Can you think of a case where a maximal solution is not complete?

Stability of bouncing ball

Simulations with $g = 9.81$, $\lambda = 0.8$, $h(0) = 1$, and $v(0) = 0$.



IV-2. Sampled-data linear systems

Sampled-data linear systems

Consider again the system

$$\begin{bmatrix} \dot{z} \\ \dot{u} \\ \dot{\tau} \end{bmatrix} = \begin{bmatrix} Az + Bu \\ 0 \\ 1 \end{bmatrix}, \text{ for } \tau \in [0, T_s]$$

$$\begin{bmatrix} z^+ \\ u^+ \\ \tau^+ \end{bmatrix} = \begin{bmatrix} z \\ Kz \\ 0 \end{bmatrix}, \text{ for } \tau = T_s$$

Define $x_1 = (z, u)$ and $x_2 = \tau$. The target set for $x = (x_1, x_2)$ is $\mathcal{A} = \{0\} \times [0, T_s]$.

Sampled-data linear systems

In terms of x_1 and x_2 we can write

$$\dot{x} = f(x) := \begin{bmatrix} A_f x_1 \\ 1 \end{bmatrix}, C := \{x : x_2 \in [0, T_s]\}$$

$$x^+ = g(x) := \begin{bmatrix} A_g x_1 \\ 0 \end{bmatrix}, D = \{x : x_2 = T_s\}$$

where

$$A_f = \begin{bmatrix} A & B \\ 0 & 0 \end{bmatrix}, A_g = \begin{bmatrix} I & 0 \\ K & 0 \end{bmatrix}.$$

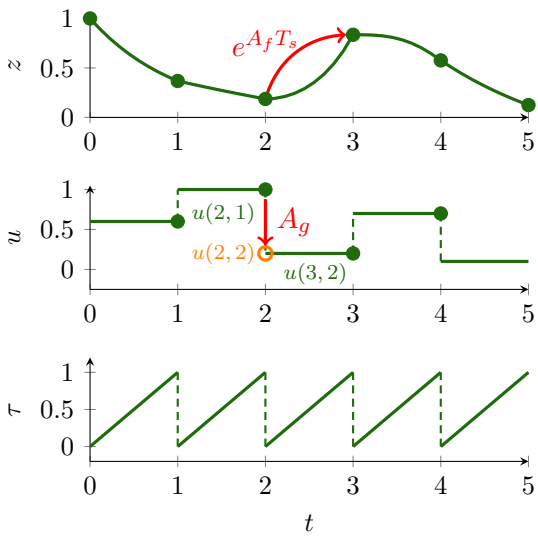
Sampled-data linear systems

Assuming $\tau(0,0) = 0$, the hybrid time domain of every complete solution ϕ is

$$\mathbf{dom}\,\phi = \bigcup_{j=0}^{\infty} [jT_s, (j+1)T_s] \times \{j\}.$$



Sampled-data linear systems



The jump at time $(j+1)T_s$ is

$$x_1((j+1)T_s, j+1) = A_g x_1((j+1)T_s, j).$$

Between two consecutive jumps,

$$x_1((j+1)T_s, j) = e^{A_f T_s} x_1(jT_s, j).$$

Therefore, defining the pre-jump state $y_j = x_1(jT_s, j-1)$ for $j \geq 1$ gives

$$y_{j+1} = e^{A_f T_s} A_g y_j = H y_j.$$

Sampled-data linear systems

Matrix H is Schur stable iff there is $P \succ 0$ such that

$$H^\intercal P H - P = -Q,$$

where $Q \succ 0$. Define $W(x_1) = x_1^\intercal P x_1$. From the discrete-time Lyapunov function W we can make a continuous-time Lyapunov function

$$V_1(x) = W(\exp(A_f(T_s - x_2))x_1),$$

which satisfies (L1) [w/b], but (L2) does not hold because [w/b]

$$\dot{V}_1(x) = 0, x \in C.$$

Condition (L3) is satisfied–there is $c > 0$ such that for $x \in D$ [w/b]

$$V_1(g(x)) - V_1(x) \leq -c\|x\|_{\mathcal{A}}^2.$$

Sampled-data linear systems

To remedy the fact that V does not decrease along the continuous-time trajectories, we introduce

$$V_2(x) = e^{-\sigma x_2} V_1(x),$$

which can be shown to be a Lyapunov function for appropriately small $\sigma > 0$.

Ergo: $\rho(H) < 1 \implies \mathcal{A} = \{0\} \times [0, T_s]$ is UGAS. If $A + BK$ is stable, the sampled-data system can be unstable (see slide 28).

Sampled-data linear systems

How to choose K. Recall $H = e^{A_f T_s} A_g$ and

$$\exp(A_f T_s) = \exp\left(\begin{bmatrix} A & B \\ 0 & 0 \end{bmatrix} T_s\right) = \begin{bmatrix} A_d & B_d \\ 0 & I \end{bmatrix},$$

where

$$A_d = e^{AT_s}, B_d = \int_0^{T_s} e^{As} B \mathrm{d}s.$$

Therefore,

$$H = \begin{bmatrix} A_d + B_d K & 0 \\ K & 0 \end{bmatrix},$$

and observe that H is block-triangular, so its spectrum is $\sigma(H) = \sigma(A_d + B_d K) \cup \{0, \ldots, 0\}$, so H is Schur stable iff $A_d + B_d K$ is Schur stable. We can determine K by pole placement (provided (A_d, B_d) is stabilisable), or by LQR.

Sampled-data linear systems: example

Consider the double-integrator plant with

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, K = \begin{bmatrix} -0.375 & -1.0625 \end{bmatrix},$$

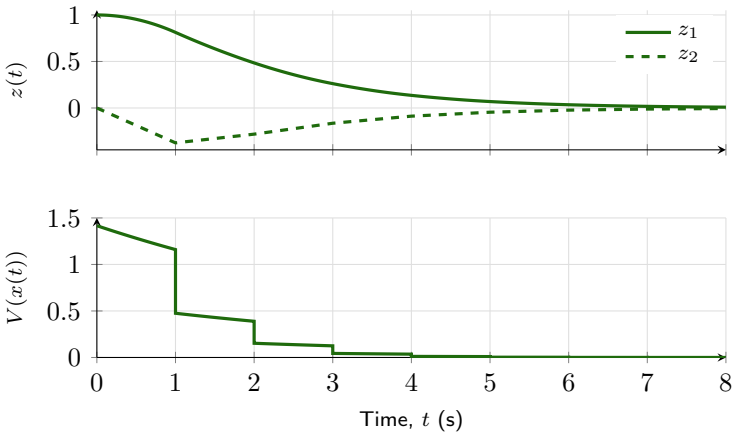
and $T_s = 1$. The pre-jump map has eigenvalues 0, 0.25, and 0.5. The solution of $H^\intercal P H - P = -I_3$ is

$$H = \begin{bmatrix} 0.8125 & 0.46875 & 0 \\ -0.375 & -0.0625 & 0 \\ -0.375 & -1.0625 & 0 \end{bmatrix}, \qquad P \approx \begin{bmatrix} 2.4167 & 1.1250 & 0 \\ 1.1250 & 2.6042 & 0 \\ 0 & 0 & 1.0000 \end{bmatrix}.$$

Since $\lambda_{\max}(P) \approx 3.6393$, choose $\sigma = 0.2 < 0.3213$ so that $(1 - e^{-\sigma T_s})\lambda_{\max}(P) < 1$. Then

$$V(x) = e^{-\sigma x_2} \| \exp(A_f(T_s - x_2)) x_1 \|_P^2.$$

Sampled-data linear systems: example



Simulation with $z(0) = (1, 0)$, $u(0) = Kz(0)$, and $x_2(0) = 0$.

IV-3. Switching systems

Switching systems

Consider the following switching system

$$\mathcal{H}: \begin{cases} \begin{cases} \dot{z} = f_q(z) \\ \dot{q} = 0 \\ \dot{\tau} \in [0, \eta] \end{cases} & \text{for } \tau \in [0, N_0] \\ \begin{cases} z^+ = z \\ q^+ \in Q \\ \tau^+ = \tau - 1 \end{cases} & \text{for } \tau \in [1, N_0] \end{cases}$$

where Q is a finite set, $\eta > 0$, $N_0 \geq 1$. There is no Zeno behaviour (why?) [w/b]

The state is $x = (z, q, \tau)$, and $C = \mathbb{R}^p \times Q \times [0, N_0]$, $D = \mathbb{R}^p \times Q \times [1, N_0]$.

The target is $\mathcal{A} = \{0\} \times Q \times [0, N_0]$.

Question: What are the set-valued functions F and G here?

Switching systems

Suppose the origin is a GES eq. point of every subsystem $\dot{z} = f_q(z)$ and let $W_q \in C^1(\mathbb{R}^p; \mathbb{R})$ such that

1. $\exists c, C \in \mathbb{R} : e^c \|z\|^2 \leq W_q(z) \leq e^C \|z\|^2$, for $z \in \mathbb{R}^p$, $q \in Q$
2. $\exists \lambda > 0 : \nabla W_q(z)^\intercal f_q(z) \leq -\lambda W_q(z)$, for $z \in \mathbb{R}^p$, $q \in Q$
3. $C - c < \lambda/\eta$

That is, we have a Lyapunov function for each subsystem. Consider the candidate Lyapunov function

$$V(x) = e^{\mu\tau} W_q(z),$$

where $\mu > 0$ is such that $C - c < \mu < \lambda/\eta$. Then V is a Lyapunov function for $\mathcal{A}$.

Switching systems

We will show that [w/b]

1. $e^c\|x\|_{\mathcal{A}}^2 \leq V(x) \leq e^{\mu N_0 + C}\|x\|_{\mathcal{A}}^2$ for all x
2. Let $F(x) = \{f_q(z)\} \times \{0\} \times [0, \eta]$; then for $f \in F(x)$ and $x \in C$ we have

$$\nabla V(x)^\intercal f \leq -\alpha\|x\|_{\mathcal{A}}^2,$$

   for a constant $\alpha > 0$.
3. It is $G(x) = \{z\} \times Q \times \{\tau - 1\}$; then, for $g \in G(x)$, and $x \in D$ we have

$$V(g) - V(x) \leq -\beta\|x\|_{\mathcal{A}}^2,$$

   for a constant $\beta > 0$.

Therefore, $\mathcal{A}$ is UGAS for the switching system.

V. Conclusions